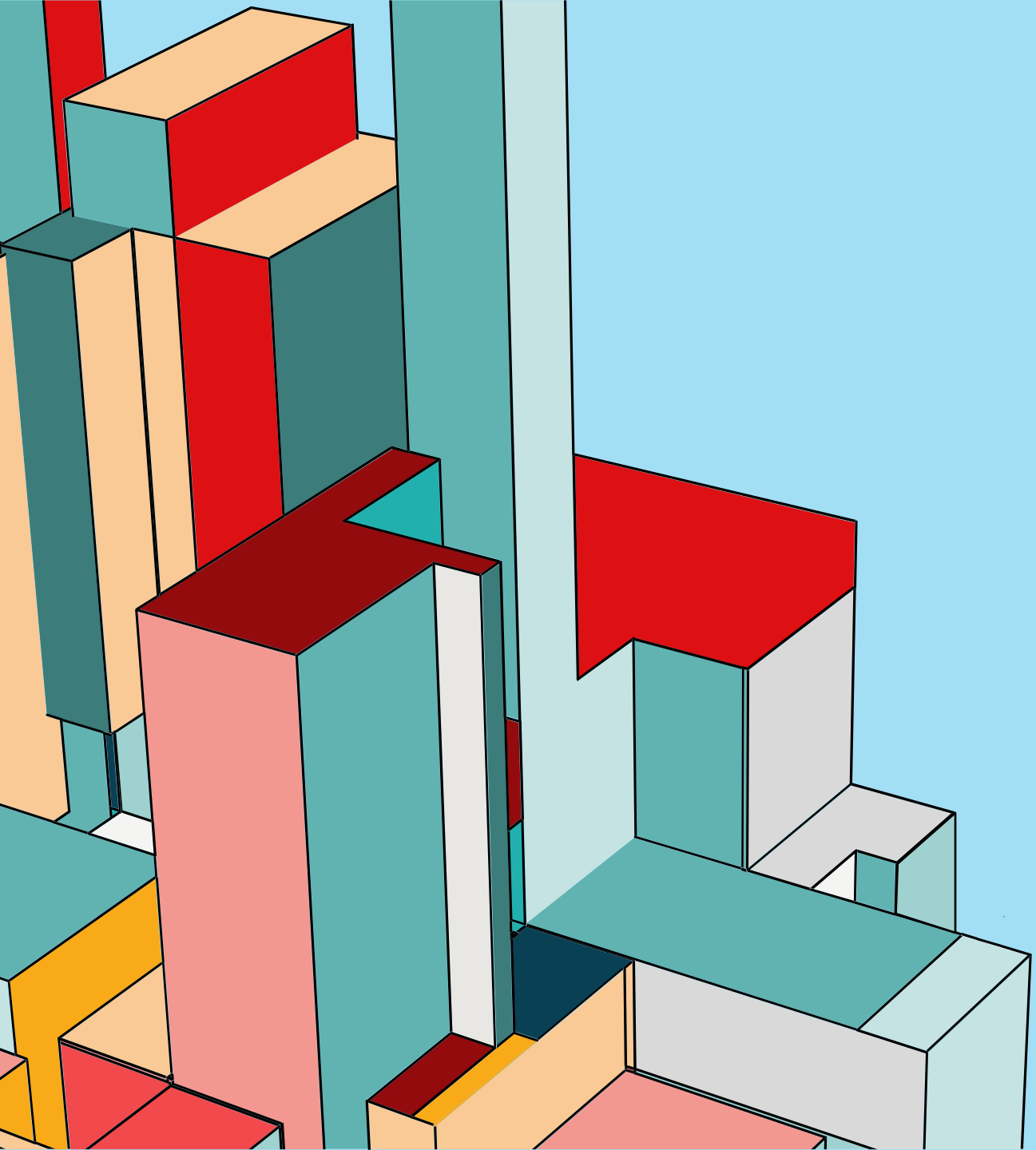


AEGIS GRID

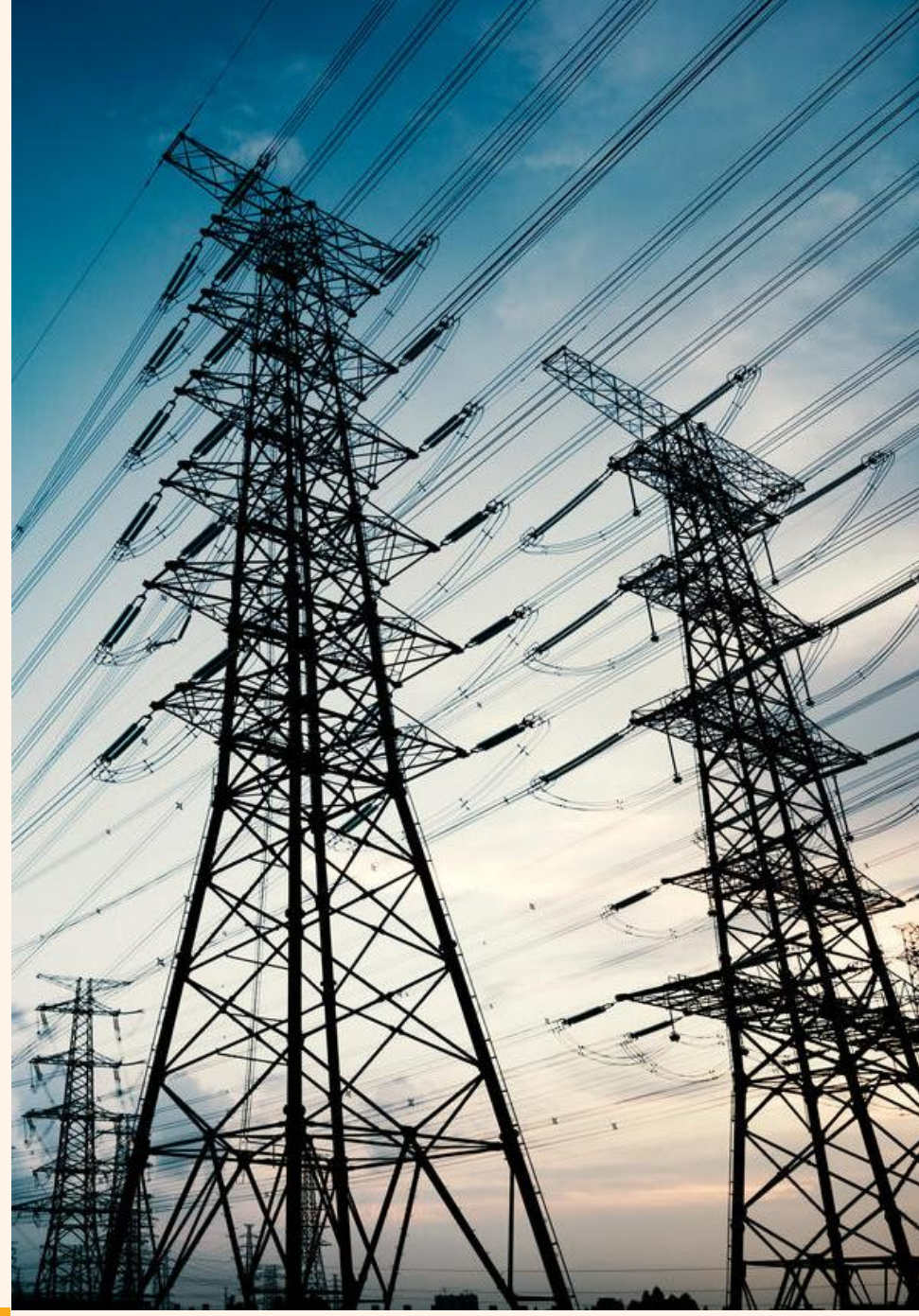
We don’t just watch the grid – we keep it current.

Brought to you by:

Nicolas Jao	400450151
Ayesha Mehmood	400403482
Yihang Qin	400461329

OUTLINE

- Problem Overview
- Why Current Systems Fail
- Stakeholders
- Proposed Solution
- Value Proposition
- Budget & Feasibility
- Proof-of-Concept



PROBLEM OVERVIEW

- According to the U.S. Department of Energy, power outages are costing American businesses around $150 billion USD per year
- According to Constellation Energy Corporation, aging infrastructure and degrading equipment is the third most common cause of power outages

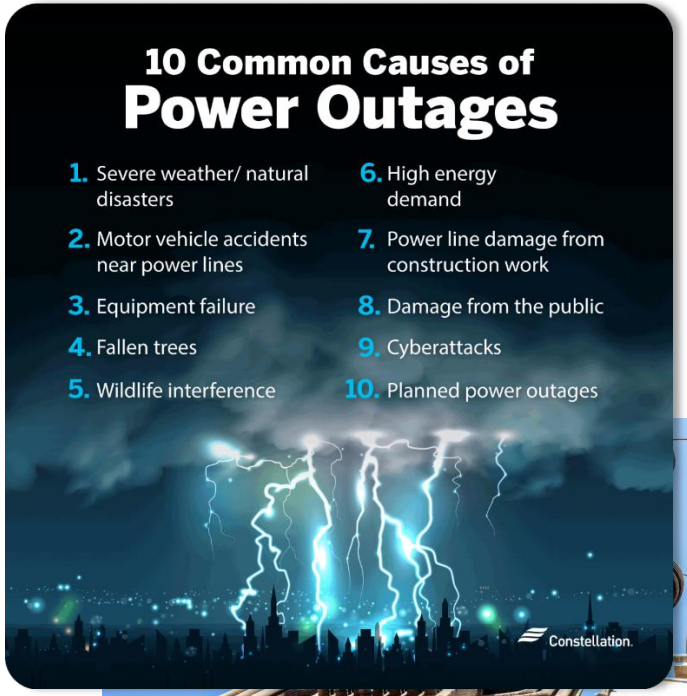


Table 3: Cost of Interruptions by Province in 2019

Province	Momentary Interruptions Cost (Billion CAD)	Sustained Interruptions Cost (Billion CAD)	Total Cost
Alberta	$0.874	$0.437	
British Columbia	$3.894	$1.947	
Manitoba	$0.719	$0.360	
New Brunswick	$4.463	$2.231	
Newfoundland and Labrador	$0.173	$0.086	
Northwest Territories	$0.140	$0.070	
Nova Scotia	$0.730	$0.365	
Nunavut	$0.011	$0.006	
Ontario	$3.655	$1.827	
Prince Edward Island	$0.036	$0.018	
Quebec	$7.223	$3.612	
Saskatchewan	$1.073	$0.536	
Yukon	$0.030	$0.015	
Total for Canada	**$23.021**	**$11.510**	**$34.530**

Note: This table excludes costs related to loss of power from major events and loss of supply.

PROBLEM OVERVIEW

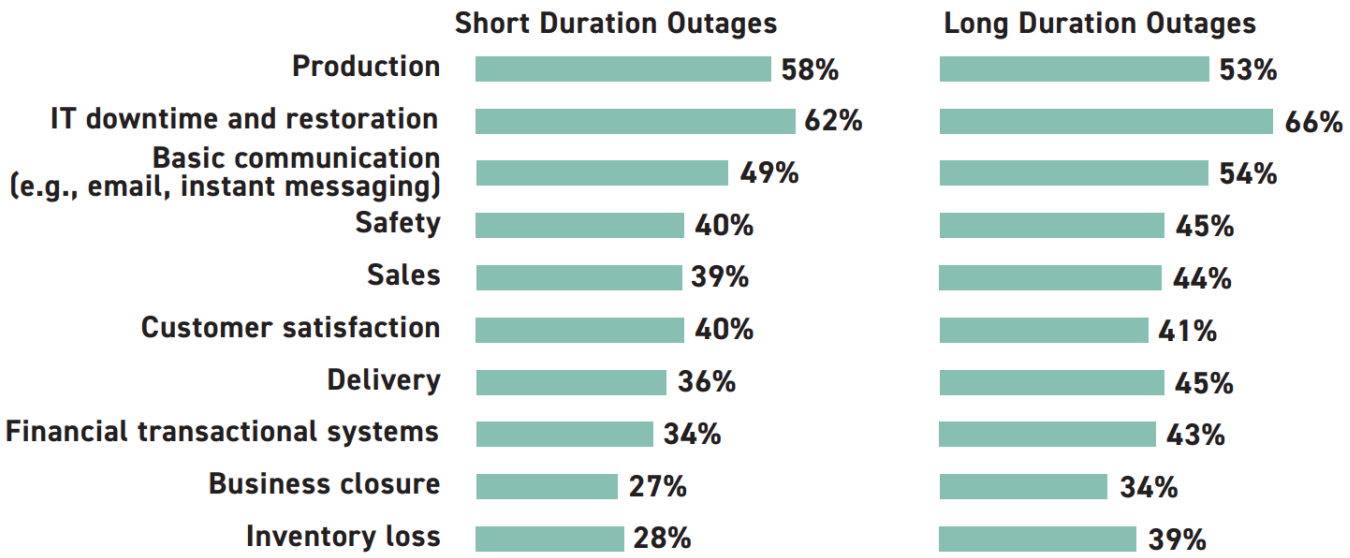
According to a study done by S&C Electric Company, in 2019, power outages for 82 Canadian utilities cost the Canadian economy $34.5 billion CAD.

That's a lot of money that could be saved for more important things!

PROBLEM OVERVIEW

- From the same report, these are all the ways businesses lose money from power interruptions or outages
- Fun fact: During my co-op, I had a few days where power was out, and I couldn't do any work the whole day!

Table 4: Impacts of Short- and Long-Duration Outage Impacts Over Past 12 Months



16-18 From S&C's "2021 State of Commercial & Industrial Power Reliability Report," March 2021.

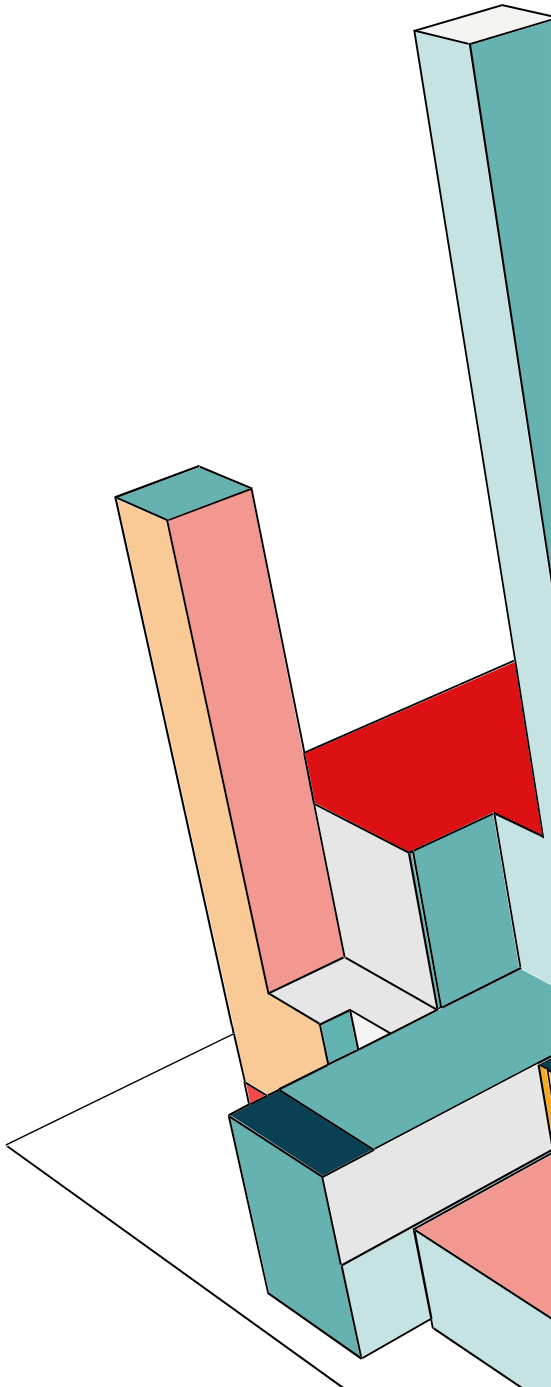
WHY CURRENT SYSTEMS FAIL

Many power grids today lack a huge thing: Data.

They lack sensors:

- At remote locations to monitor signs of high-power demand
- On transmission lines to detect locations of faults when blackouts occur
- On equipment to detect signs of failure before they happen

These issues cause more frequent outages, longer outages, and higher maintenance costs!



WHY CURRENT SYSTEMS FAIL

Legacy Systems

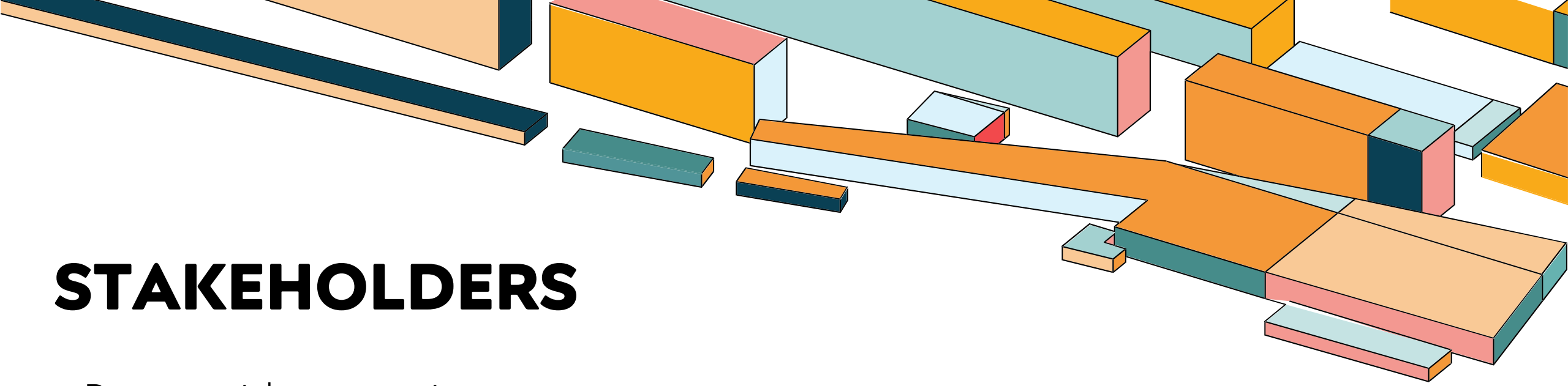
- SCADA and RTUs only measure voltage/current at substations
- Lack of fault location data, utilities wastes time finding them

Reactive Maintenance

- Equipment degradation is detected after failure
- Lack of real-time data on equipment

Limited Data Collection

- Remote areas are under-monitored
- Lack of data collection range



STAKEHOLDERS

- Power grid companies
- Municipal, provincial, federal governments
- Emergency services and hospitals
- Airports
- Homeowners
- C&I Businesses (factories, offices, schools, warehouses)
- Everyone who uses electricity in towns and cities, including you!

That’s a lot of people affected!

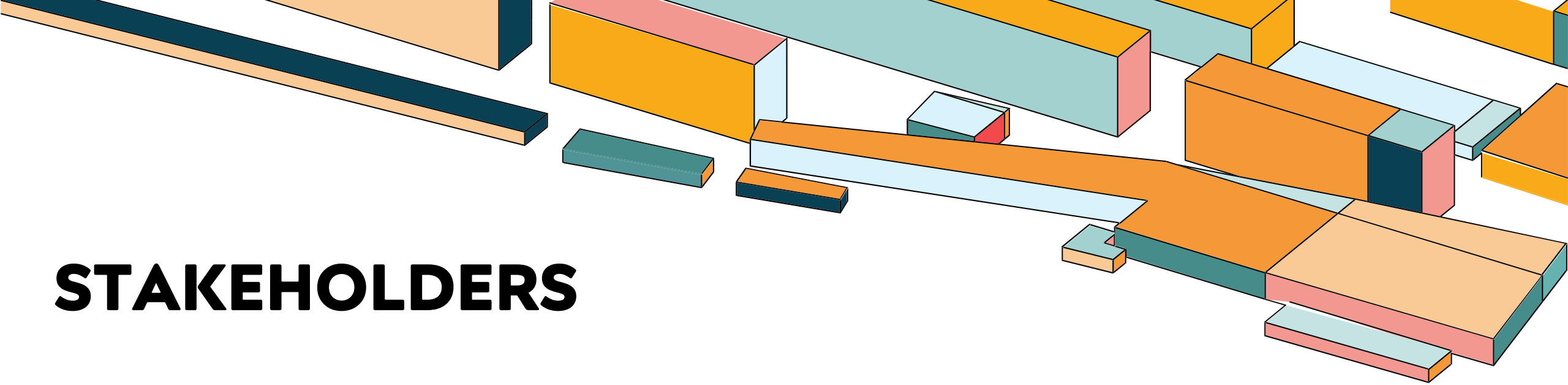
Appendix B: List of All Utilities Considered for this Study

Utility Name	Province/ Territory	Utility Name	Province/ Territory
Alectra Utilities Corporation	Ontario	Oakville Hydro Electricity Distribution Inc.	Ontario
Algoma Power Inc.	Ontario	Orangeville Hydro Limited	Ontario
Atikokan Hydro Inc.	Ontario	Orillia Power Distribution Corporation	Ontario
Bluewater Power Distribution Corporation	Ontario	Oshawa River Power Corporation	Ontario
Brantford Power Inc.	Ontario	Ottawa River Power Corporation	Ontario
Burlington Hydro Inc.	Ontario	Peterborough Distribution Incorporated	Ontario
Canadian Niagara Power Inc.	Ontario	PUC Distribution Inc.	Ontario
Centre Wellington Hydro Ltd.	Ontario	Renfrew Hydro Inc.	Ontario
Chapleau Public Utilities Corporation.	Ontario	Rideau St. Lawrence Distribution Inc.	Ontario
Cooperative Hydro Embrun Inc.	Ontario	Sioux Lookout Hydro Inc.	Ontario
E.L.K. Energy Inc.	Ontario	Synergy North Corporation	Ontario
Elexicon Energy Inc.	Ontario	Tillsonburg Hydro Inc.	Ontario
Energy+ Inc.	Ontario	Toronto Hydro-Electric System Limited	Ontario
Entegrus Powerlines Inc.	Ontario	Wasaga Distribution Inc.	Ontario
ENWIN Utilities Ltd.	Ontario	Waterloo North Hydro Inc.	Ontario
EPCOR Electricity Distribution Ontario	Ontario	Welland Hydro-Electric System Corp.	Ontario
ERTH Power Corporation	Ontario	Wellington North Power Inc.	Ontario
Espanola Regional Hydro Distribution	Ontario	Westario Power Inc.	Ontario
Essex Powerlines Corporation	Ontario	ATCO Electric	Alberta
Festival Hydro Inc.	Ontario	City of Lethbridge	Alberta
Fort Frances Power Corporation	Ontario	City of Medicine Hat	Alberta
Greater Sudbury Hydro Inc.	Ontario	City of Red Deer	Alberta
Grimsby Power Inc.	Ontario	ENMAX	Alberta
Halton Hills Hydro Inc.	Ontario	EPCOR	Alberta
Hearst Power Distribution Company Ltd.	Ontario	Fortis Alberta	Alberta
Hydro 2000 Inc.	Ontario	Manitoba Hydro	Manitoba
Hydro Hawkesbury Inc.	Ontario	BC Hydro	British Columbia
Hydro One Networks Inc.	Ontario	FortisBC	British Columbia
Hydro Ottawa Limited	Ontario	Hydro Quebec	Quebec
Innpower Corporation	Ontario	Nova Scotia Power	Nova Scotia
Kingston Hydro Corporation	Ontario	New Brunswick Power	New Brunswick
Kitchener-Wilmot Hydro Inc.	Ontario	St. John Energy	New Brunswick
Lakefront Utilities Inc.	Ontario	Newfoundland Power	Newfoundland & Labrador
Lakeland Power Distribution Ltd.	Ontario	Newfoundland & Labrador Hydro	Newfoundland & Labrador
London Hydro Inc.	Ontario	SaskPower	Saskatchewan
Milton Hydro Distribution Inc.	Ontario	Saskatoon Light & Power	Saskatchewan
Newmarket-Tay Power Distribution Ltd.	Ontario	Northwest Territories Power Corporation	Northwest Territories
Niagara Peninsula Energy Inc.	Ontario	Maritime Electric	Prince Edward Island
Niagara-on-the-Lake Hydro Inc.	Ontario	Qulliq Energy	Nunavut
North Bay Hydro Distribution Limited	Ontario	Yukon Energy	Yukon
Northern Ontario Wires Inc.	Ontario	ATCO Electric Yukon	Yukon

STAKEHOLDERS

From the same S&C Electric report, this is a list of all utilities considered for this study.

That's a lot of potential customers for Aegis Grid!



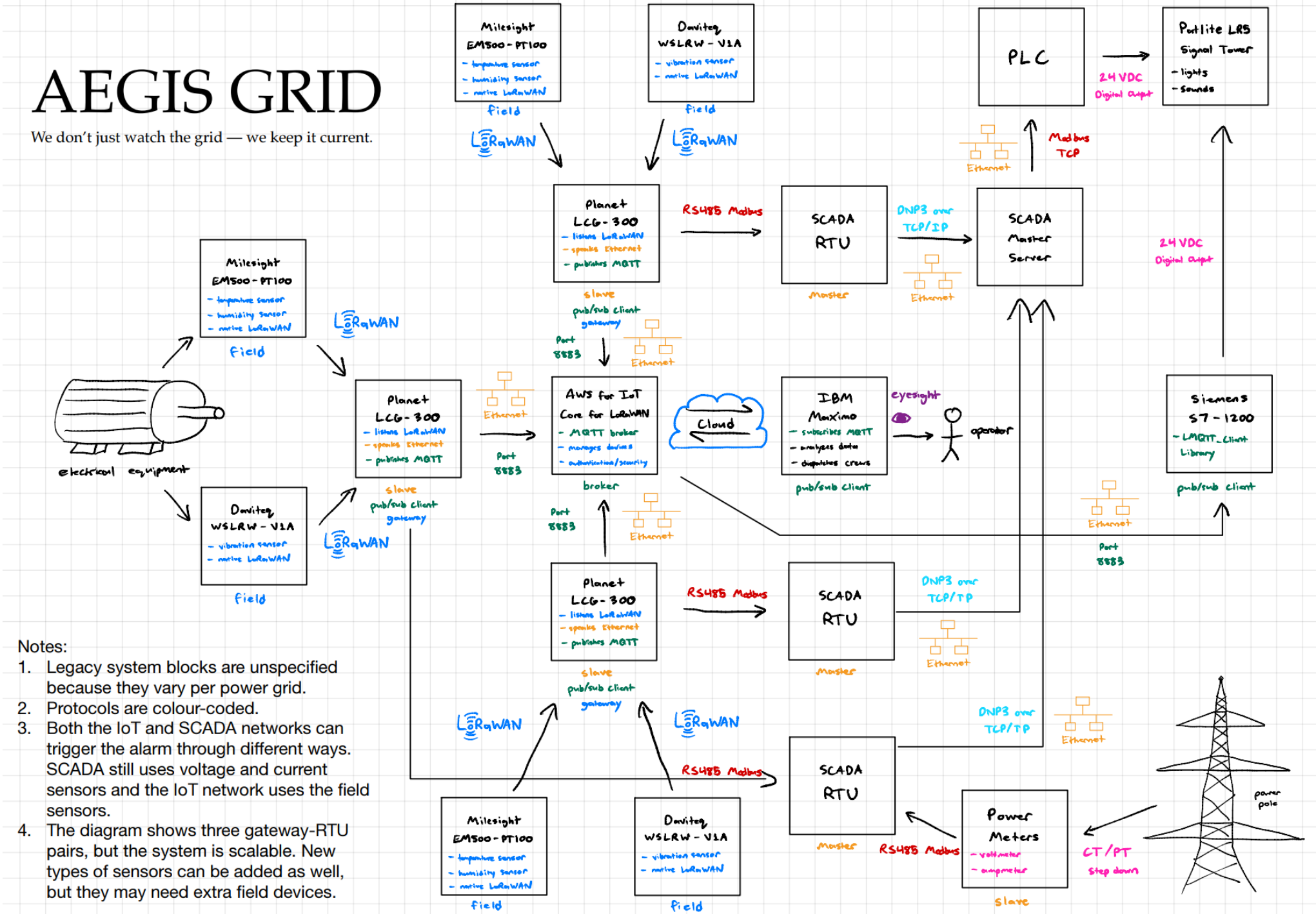
STAKEHOLDERS

What do they need or expect from a solution?

- Reduced or eliminated outage times
- As cost-effective as possible
- Can be physically implemented anywhere
- Data is reliable and accessible from anywhere in real-time

These could be solved by an IoT-based solution!

PROPOSED SOLUTION

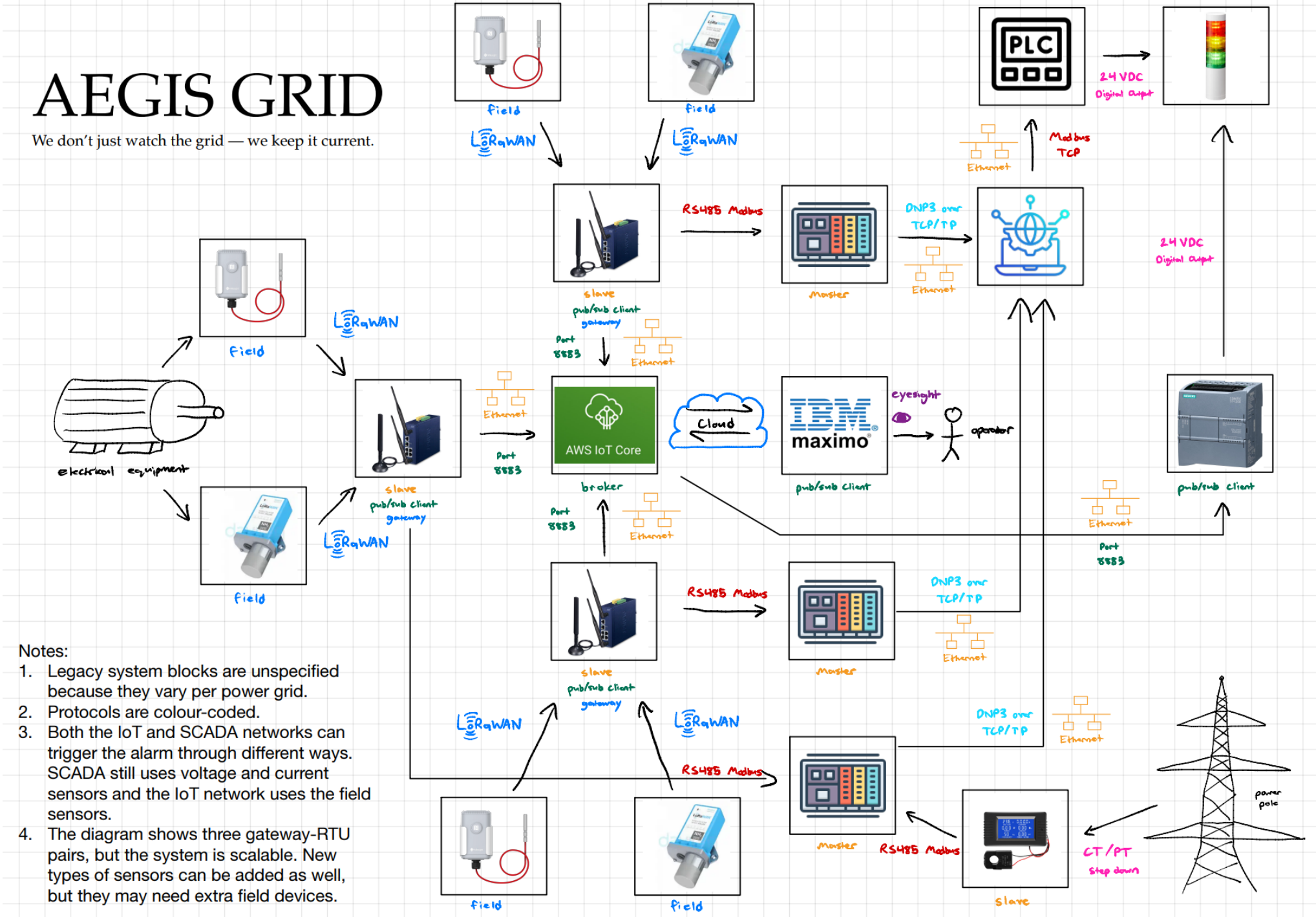


Notes:

1. Legacy system blocks are unspecified because they vary per power grid.
2. Protocols are colour-coded.
3. Both the IoT and SCADA networks can trigger the alarm through different ways. SCADA still uses voltage and current sensors and the IoT network uses the field sensors.
4. The diagram shows three gateway-RTU pairs, but the system is scalable. New types of sensors can be added as well, but they may need extra field devices.

PROPOSED SOLUTION

WITH PICTURES



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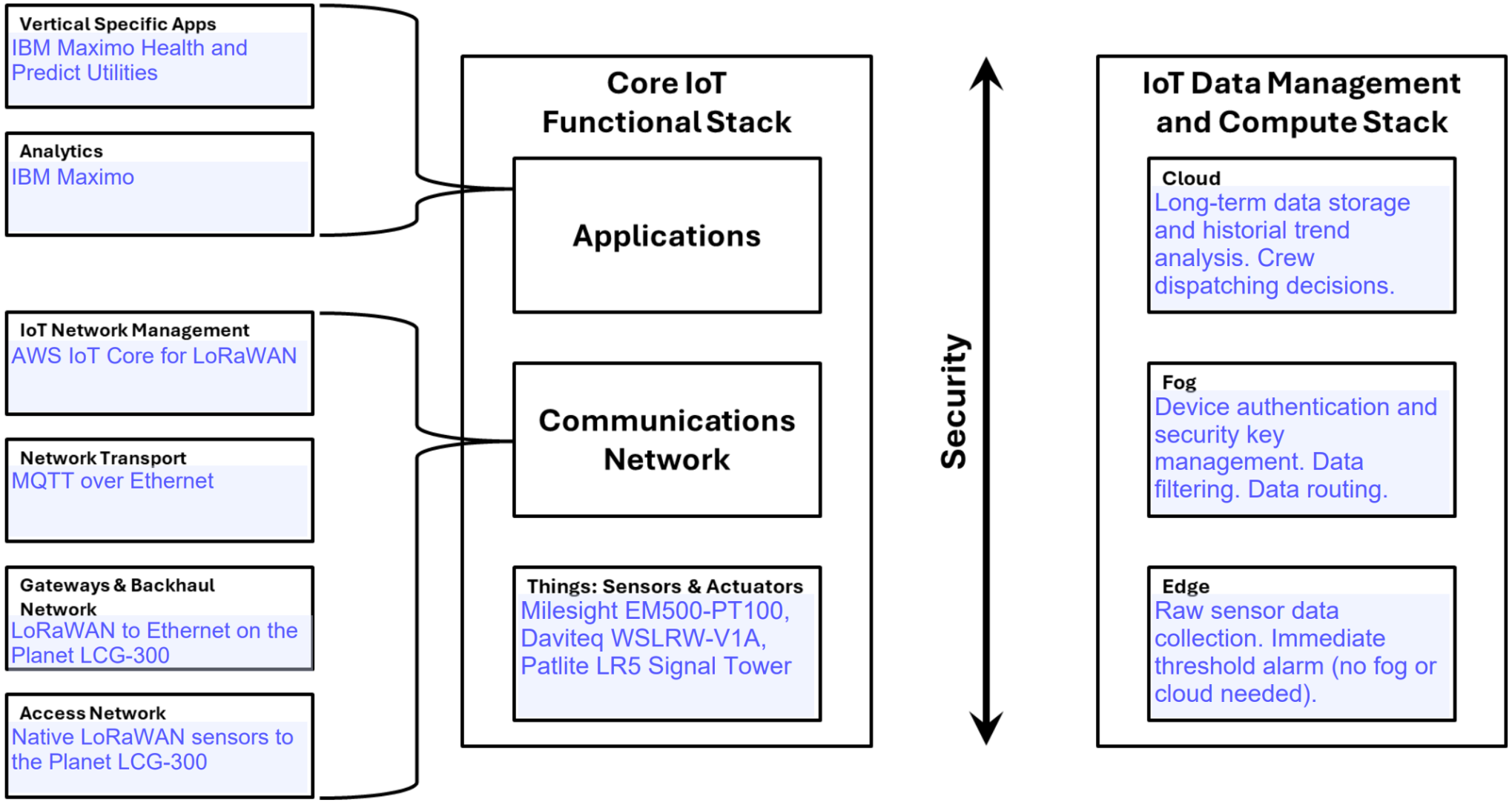
PROPOSED SOLUTION

SYSTEM ARCHITECTURE



PROPOSED SOLUTION

COMPONENT SELECTION JUSTIFICATION

Milesight EM500-PT100
- Industrial-grade
- 10 year battery life
- For harsh outdoor environments
- Native LoRaWAN support

Daviteq WSLRW-V1A
- Industrial-grade
- Frequency response of over 10kHz for bearing wear and mechanical imbalance
- Native LoRaWAN support

Planet LCG-300
- One of the only gateways that speaks LoRaWAN on the sensor side and both MQTT over Ethernet and Modbus TCP on the network side for legacy systems such as SCADA and PLCs

Patlite LR5 Signal Tower
- Standard lights and sounds for an alarm signal (could pick any)

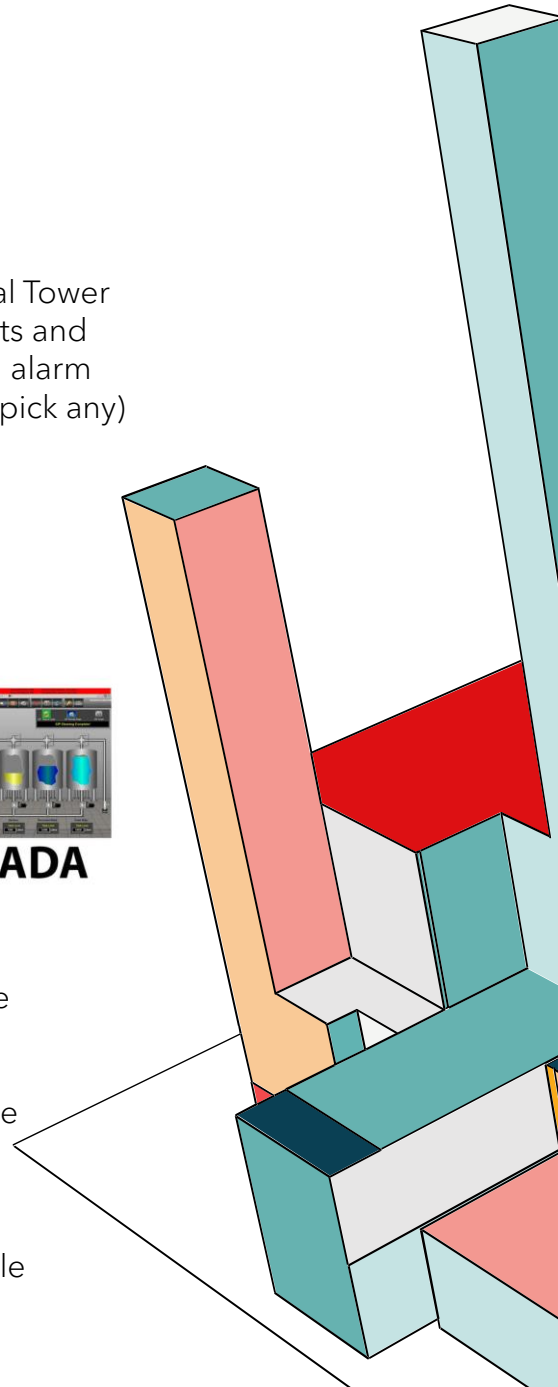
LoRaWAN
- Power grids span large geographic areas, and it is long range
- Wide range of sensors may be incompatible with the same field/gateway devices, and it is point to multipoint

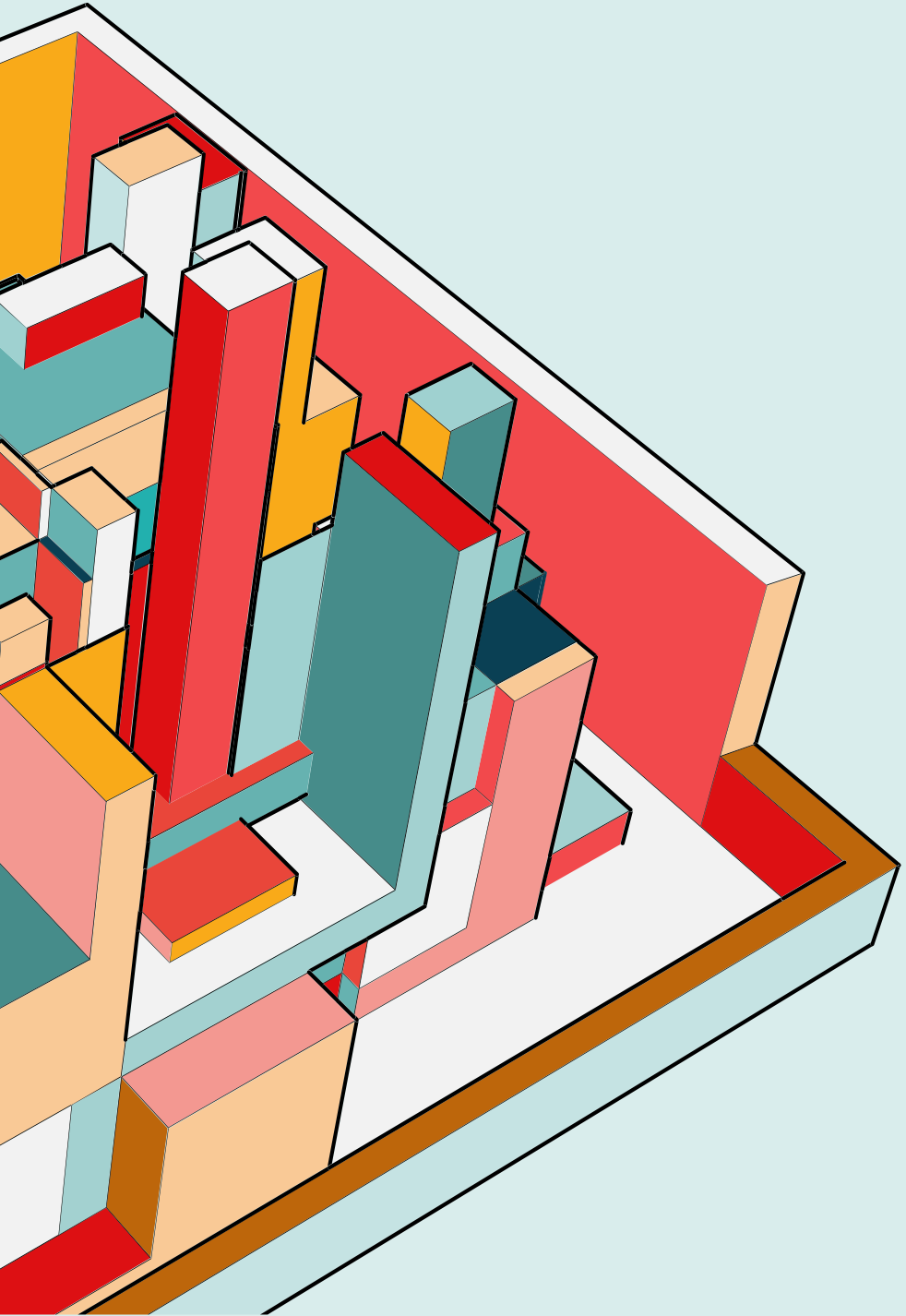
MQTT
- Lightweight
- Efficient
- Reliable
- De facto standard for IoT networks

AWS IoT Core For LoRaWAN
- A fully managed service for a LoRaWAN network

IBM Maximo
- A dominant platform for IoT analytics for utilities

Legacy Hardware/Software
- Not part of the IoT network but retained because power grids already run on them
- Extremely reliable





VALUE PROPOSITION

The core problem being solved is information latency and deficiency in the power grid.

Right now, utilities are running blind in many areas—they find out something is wrong only after it has already failed. The value of this system shifts the reactive to predictive.

BUDGET & FEASIBILITY

Hardware Components

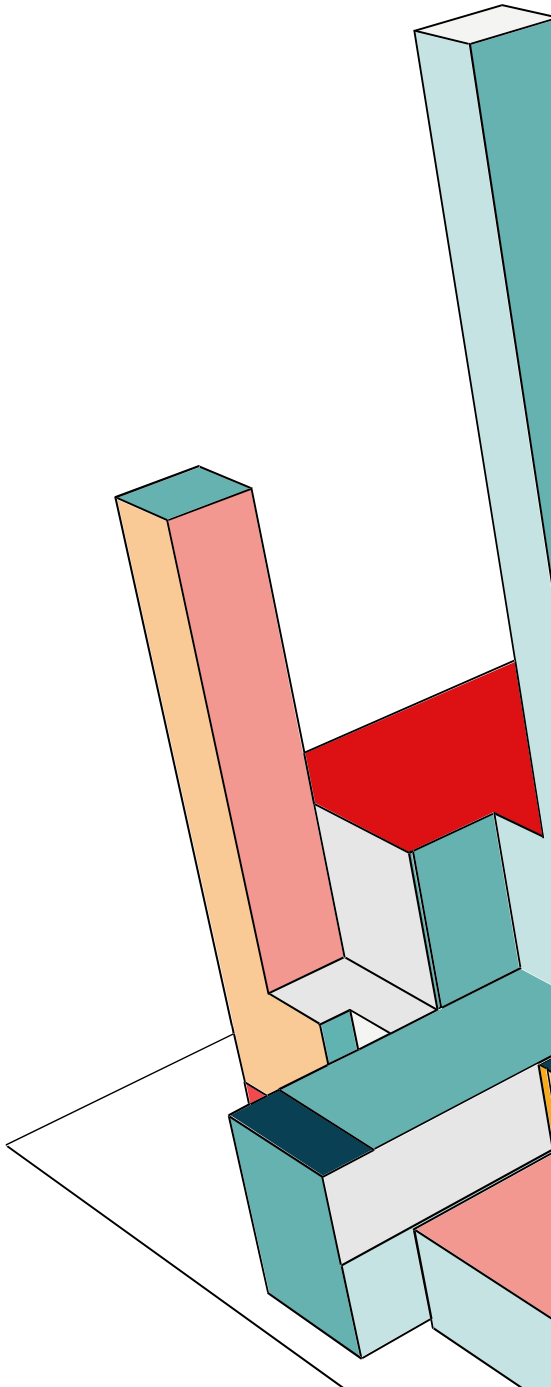
- Milesight EM500-PT100 sensors (x5)
  - $170 each -> Total $850
- Planet LCG-300 gateway
  - Estimated $500 - $800
- Daviteq WSLRW-V1A sensors (x5)
  - $150 each -> Total $750
- Patlite LR5 tower light (x1)
  - $200 each -> Total $200

Etc…

Total Estimated Base Hardware Cost: $2,300 – $2,600 per deployment unit

Cost Justification

- Sensor deployment is a variable cost based on required grid coverage
- Overall cost is low compared to outage losses (millions per incident)
- Supports proactive maintenance, reducing long-term operational costs



BUDGET & FEASIBILITY

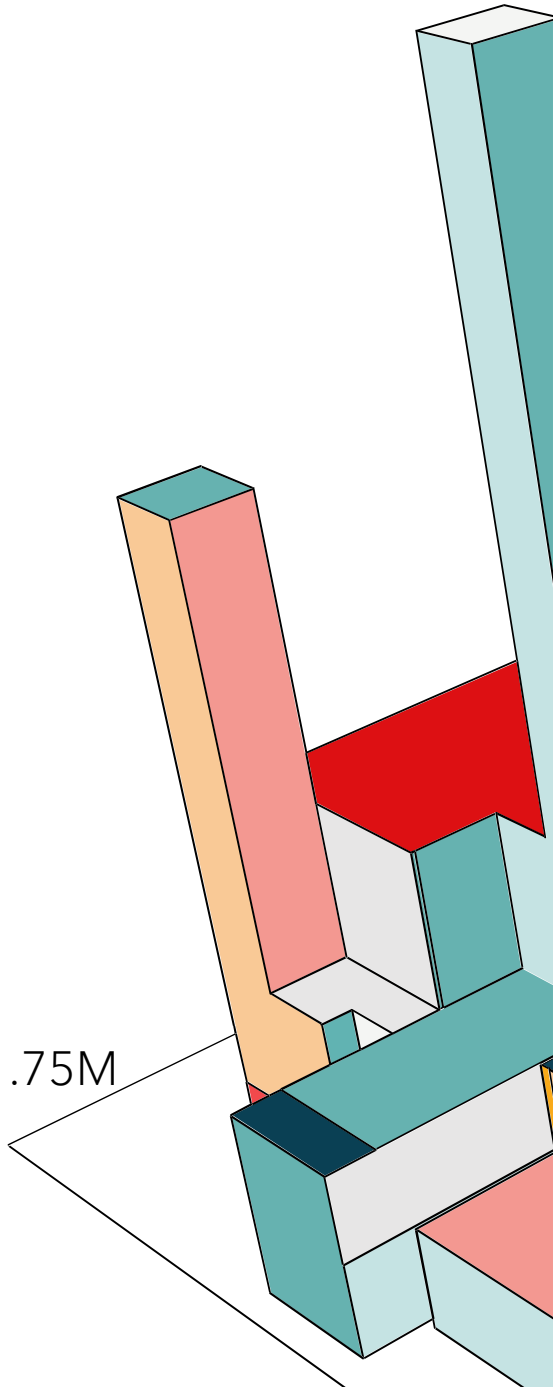
Software Components

- AWS IoT Core for LoRaWAN
  - $4/year per site
- IBM Maximo Health and Predict Utilities
  - $50,000 - $150,000+ per year licensing

Total Estimated Base Software Cost: ~$105,000 – 360,000 for Year 1
Ongoing annual: ~$100,000 - $350,000

Cost Justification

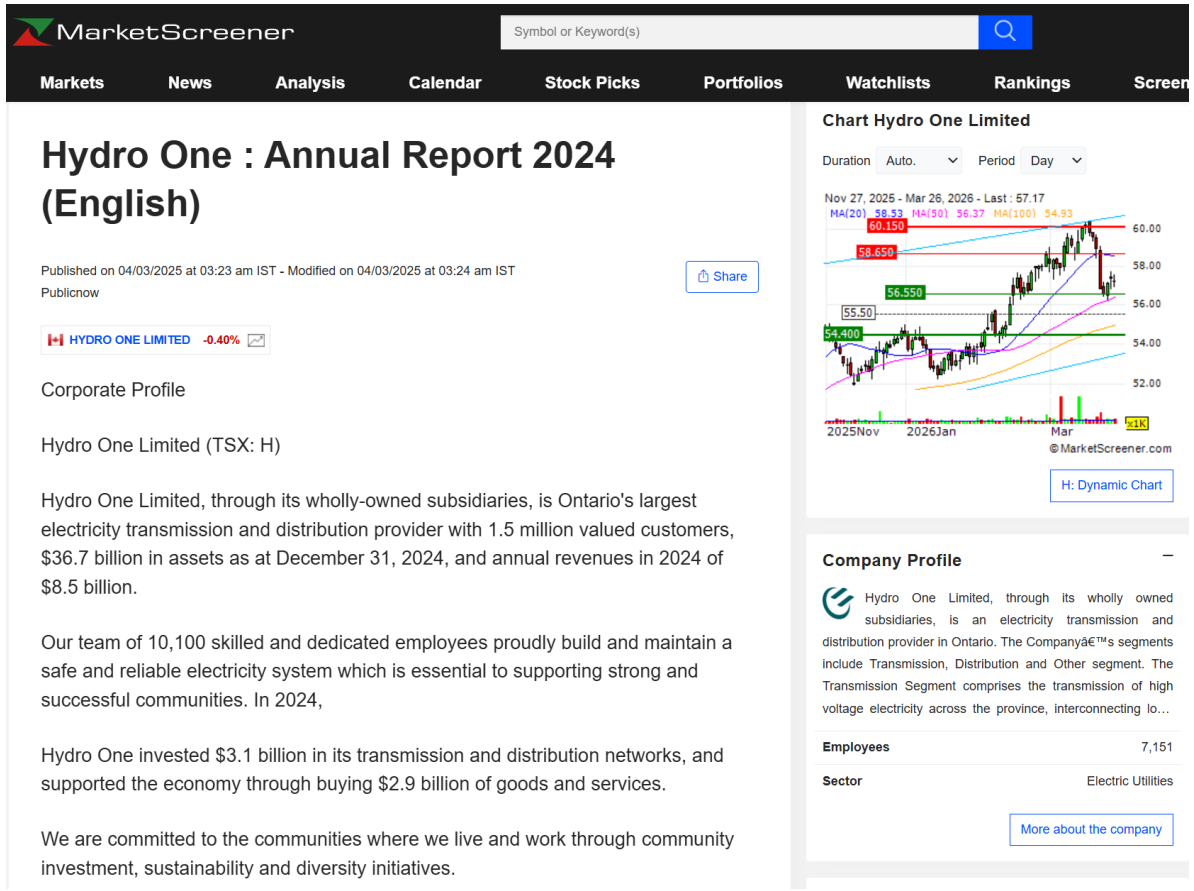
- Software licensing is a fixed cost that doesn't change per year
- Software costs do not increase as the system is scaled
- For AWS: 10 sensors × 480 messages/day × 365 days = ~1.75M messages/year, 1.75M × $2.30/million = ~$4/site × 10 sites = ~$40/year (essentially negligible)



BUDGET & FEASIBILITY

Do power utilities have the money to implement this solution?

- According to MarketScreener, Hydro One had $36.7 billion in assets and $8.5 billion in revenues in 2024
- We believe Aegis Grid's ROI is very worth it, as costs are negligible to the revenues of large power companies



MarketScreener

Symbol or Keyword(s)

Markets | News | Analysis | Calendar | Stock Picks | Portfolios | Watchlists | Rankings | Screen

Hydro One : Annual Report 2024 (English)

Published on 04/03/2025 at 03:23 am IST - Modified on 04/03/2025 at 03:24 am IST
Publicnow

Share

HYDRO ONE LIMITED -0.40%

Corporate Profile

Hydro One Limited (TSX: H)

Hydro One Limited, through its wholly-owned subsidiaries, is Ontario's largest electricity transmission and distribution provider with 1.5 million valued customers, $36.7 billion in assets as at December 31, 2024, and annual revenues in 2024 of $8.5 billion.

Our team of 10,100 skilled and dedicated employees proudly build and maintain a safe and reliable electricity system which is essential to supporting strong and successful communities. In 2024,

Hydro One invested $3.1 billion in its transmission and distribution networks, and supported the economy through buying $2.9 billion of goods and services.

We are committed to the communities where we live and work through community investment, sustainability and diversity initiatives.

Chart Hydro One Limited

Duration Auto. Period Day

Nov 27, 2025 - Mar 26, 2026 - Last : 57.17
MA(20) 58.53 MA(50) 56.37 MA(100) 54.93
60.150 58.650 56.550 55.50 54.400
60.00 58.00 56.00 54.00 52.00
2025Nov 2026Jan Mar
×1K
©MarketScreener.com

H: Dynamic Chart

Company Profile

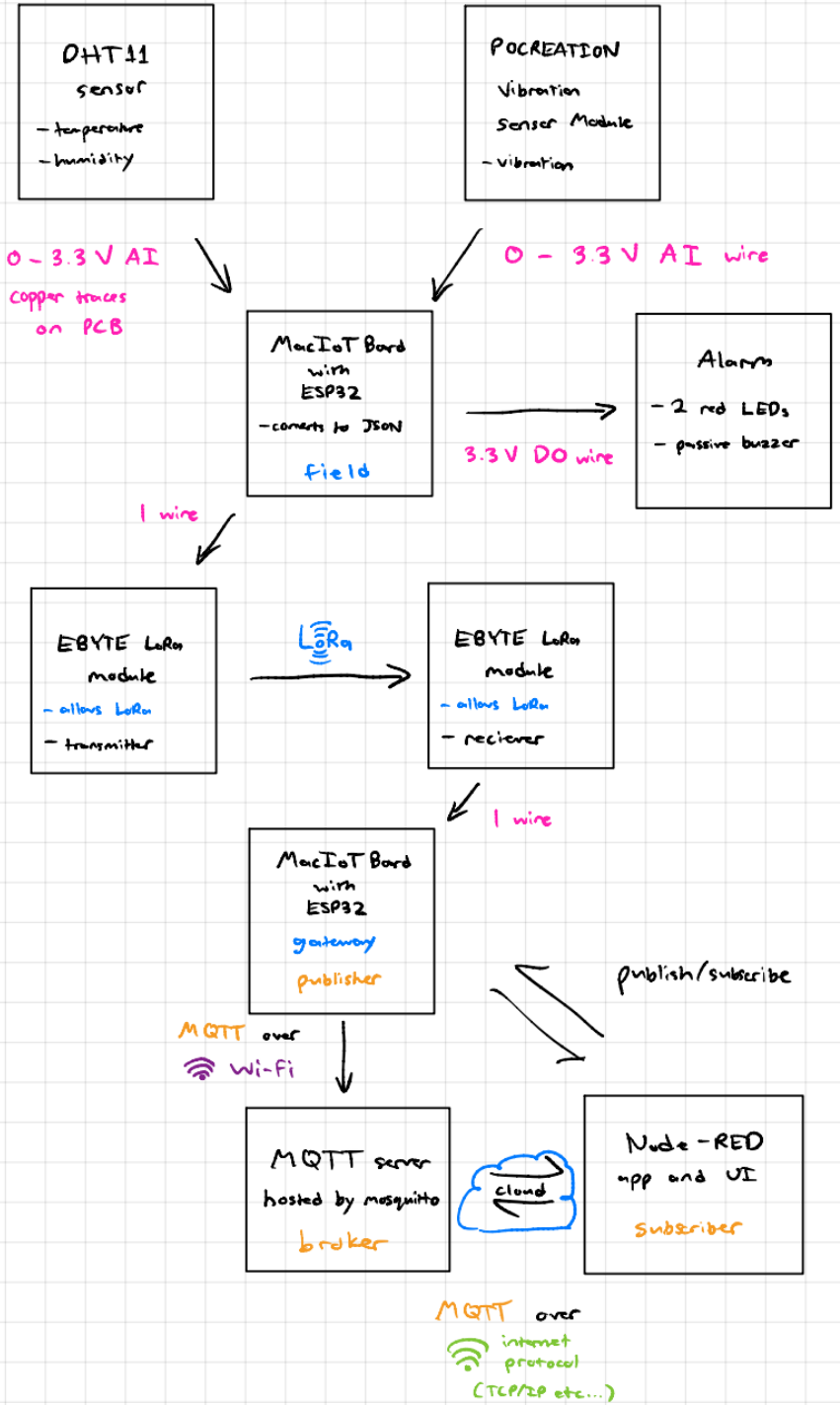
Hydro One Limited, through its wholly owned subsidiaries, is an electricity transmission and distribution provider in Ontario. The Companyâ€™s segments include Transmission, Distribution and Other segment. The Transmission Segment comprises the transmission of high voltage electricity across the province, interconnecting lo…

Employees	7,151
Sector	Electric Utilities

More about the company

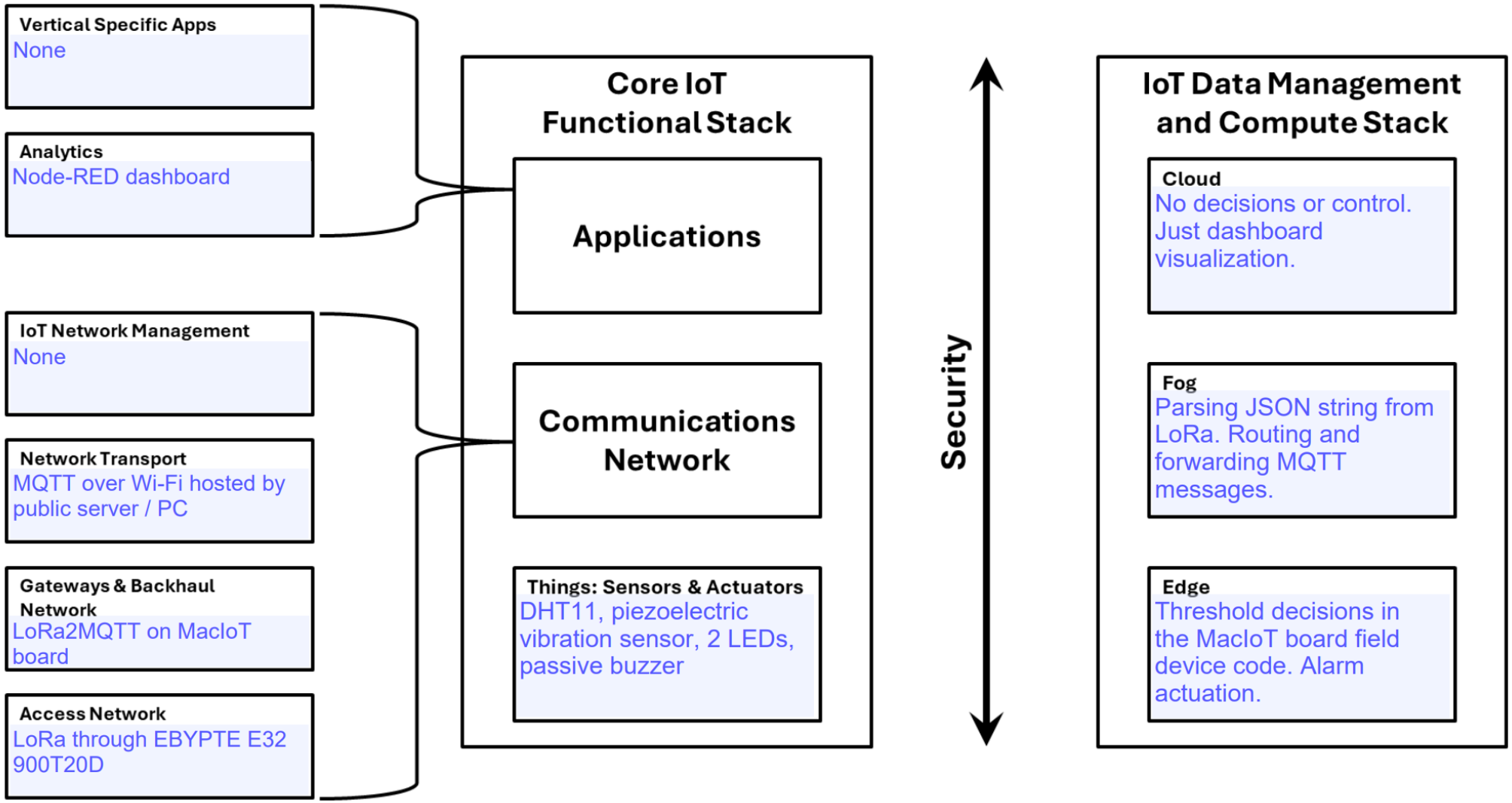
PROOF-OF-CONCEPT

AEGIS GRID proof of concept



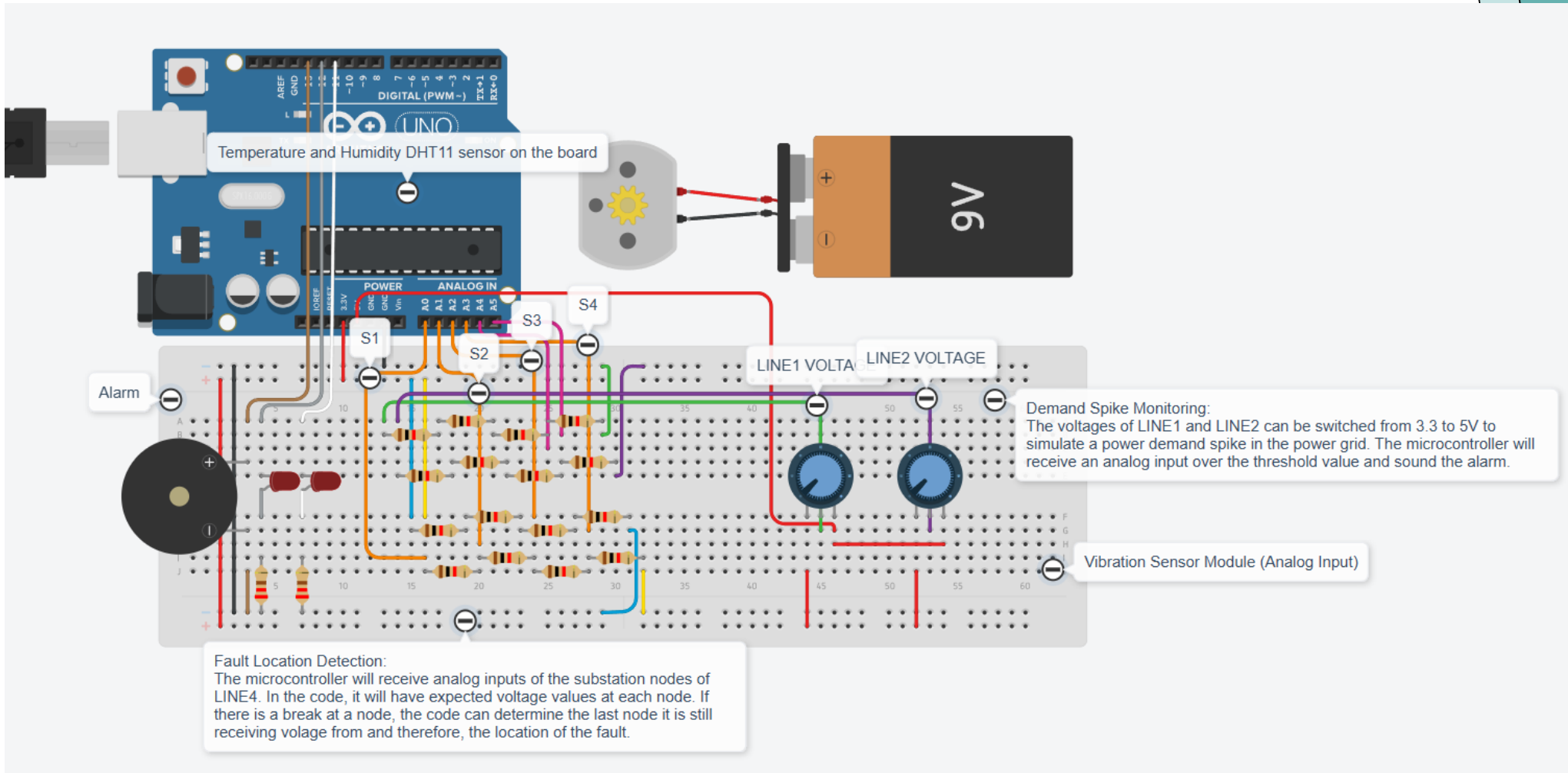
PROOF-OF-CONCEPT

SYSTEM ARCHITECTURE



PROOF-OF-CONCEPT

SCHEMATIC



PROOF-OF-CONCEPT

BOM

Qty	Description	Purchase link	Price
2	MacIoT board	register usage	On Hand
1	DHT11 temperature & humidity sensor	register usage	On Hand
2	EBYTE LoRa module	register usage	On Hand
1	Piezoelectric ceramic vibration sensor module	Link	$7.00
2	Red LED	On Hand	On Hand
1	passive buzzer	On Hand	On Hand
2	potentiometer	On Hand	On Hand
1	DC motor	On Hand	On Hand
1	9V battery	On Hand	On Hand
2	breadboard	On Hand	On Hand
16	1k resistor	On Hand	On Hand
2	220 resistor	On Hand	On Hand
a bunch	wires	On Hand	On Hand
		Total:	$7

PROOF-OF-CONCEPT

COMPONENT COMPARISON WITH PROPOSED SOLUTION

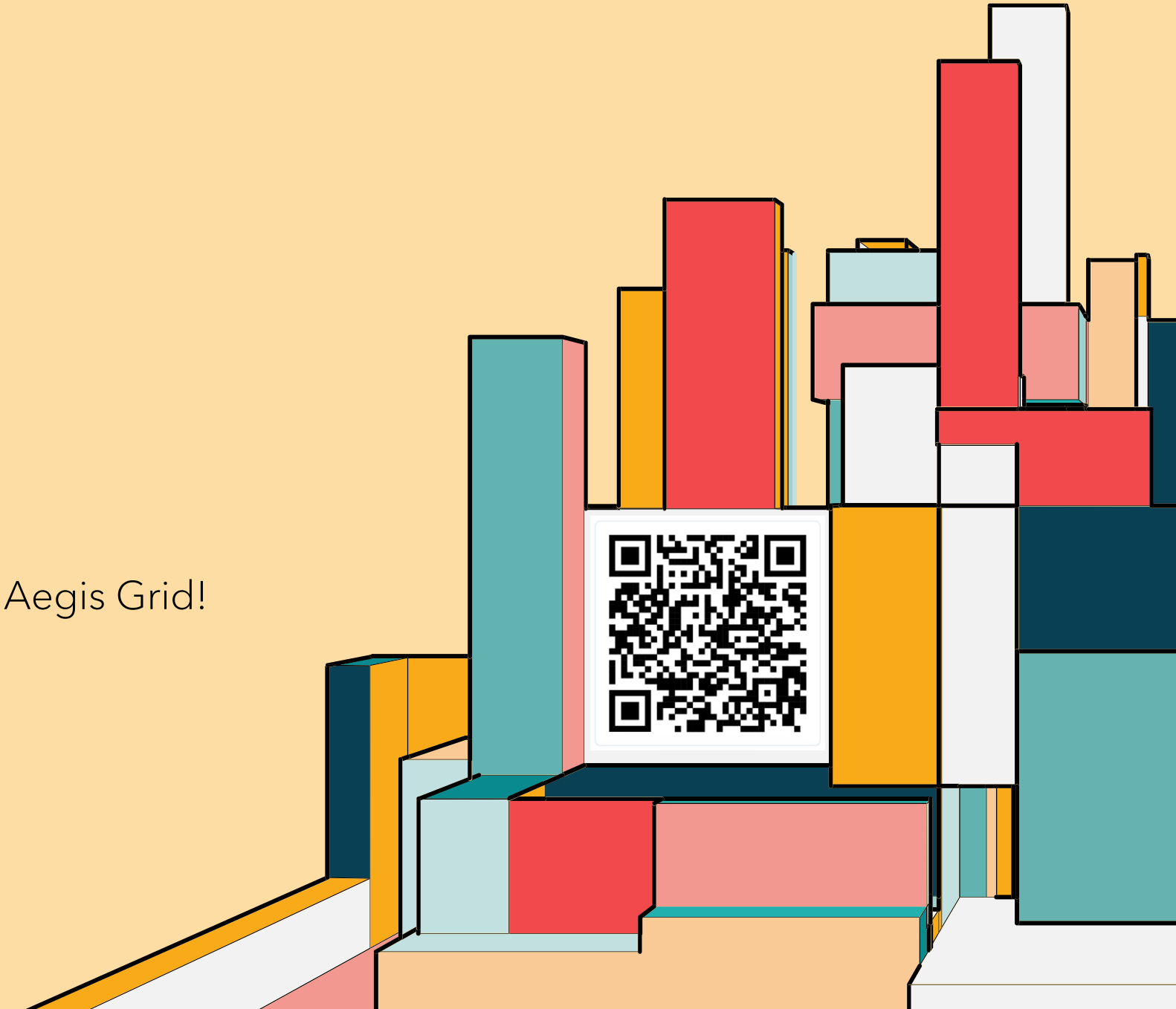
Purpose	Proof-of-Concept Component	Industrial Solution Component
Field Device	MacIoT board	None
Gateway Device	MacIoT board	Planet LCG-300
LoRa Module	EBYTE LoRa module	None
Temperature/Humidity Sensor	DHT11 temperature & humidity sensor	Milesight EM500-PT100
Vibration Sensor	Piezoelectric ceramic vibration sensor module	Daviteq WSLRW-V1A
Alarm	Red LEDs Passive Buzzers	Patlite LR5 Signal Tower
Application Software	None	IBM Maximo Health and Predict Utilities
Dashboard	Node-RED	IBM Maximo Health and Predict Utilities
IoT Network Management	None	AWS IoT Core for LoRaWAN
Voltage/Current Sensors	MacIoT board Analog Pins	CT/PT Transformer Step-Down to Meters
Network Access Protocol	LoRa	LoRaWAN
Network Transport	MQTT over Wi-Fi	MQTT over Ethernet
Power Grid Equipment	DC Motor	Generators, Transformers, Transmission Lines, etc.

Differences With Industrial Solution:

- All alarm decision-making is done at the edge, on the field device code
- The cloud is only visualization, no analytics or decision-making
- LoRa for simplicity because all our sensors interface with the MacIoT board
- MQTT is published over Wi-Fi, not Ethernet
- No vertical specific app or IoT network management software
- Has a field device (custom PCB with ESP32 microcontroller)
- Has a separate EBYTE LoRa module that the sensors interface into the GPIO

PROOF-OF-CONCEPT

DEMO VIDEO

THANKS FOR LISTENING

Be one of the first investors into Aegis Grid!